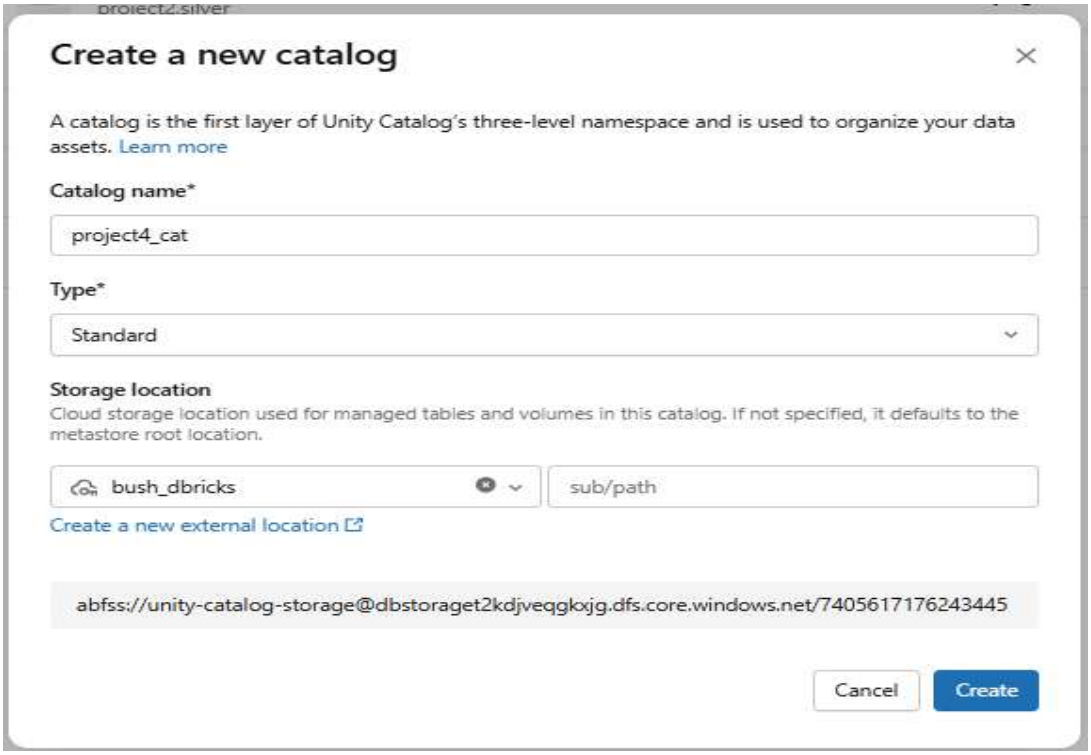
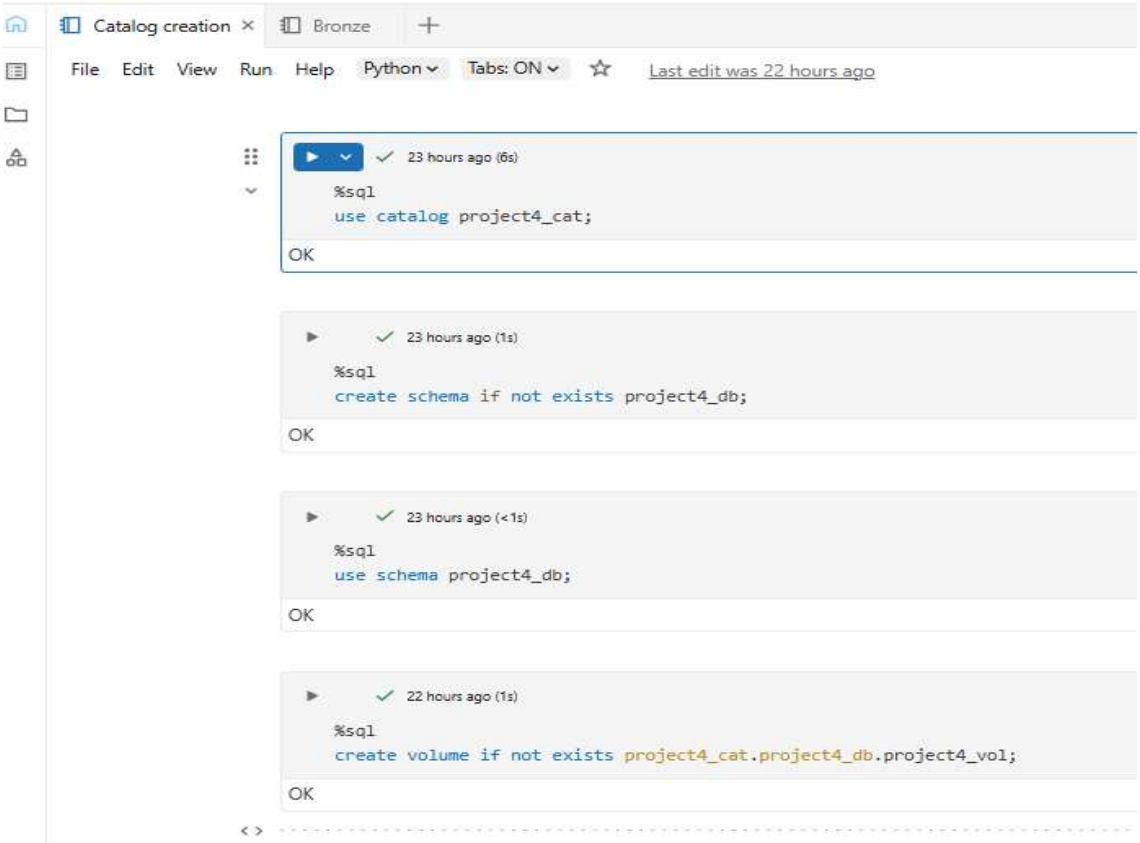
Project 4:
Catalog, schema and volume creation.





Source to Bronze layer:

24 hours ago (7s) 1

```python
import requests

# Base URL (static part)
base_url = "https://raw.githubusercontent.com/MicrosoftLearning/mslearn-databricks/main/data/"

# Dynamic years list
years = [2019, 2020, 2021]

# Target directory (Databricks volume)
raw_dir = "/Volumes/project4_cat/project4_db/project4_vol"

for year in years:
    url = f"{base_url}{year}.csv"
    print(f"Downloading {url}")

    response = requests.get(url)

    if response.status_code == 200:
        with open(f"{raw_dir}/{year}.csv", "wb") as f:
            f.write(response.content)
    else:
        print(f"Failed to download {year}")

# List files
display(dbutils.fs.ls(raw_dir))
```

(2) Spark Jobs

```
Downloading https://raw.githubusercontent.com/MicrosoftLearning/mslearn-databricks/main/data/2019.csv
Downloading https://raw.githubusercontent.com/MicrosoftLearning/mslearn-databricks/main/data/2020.csv
Downloading https://raw.githubusercontent.com/MicrosoftLearning/mslearn-databricks/main/data/2021.csv
```

Table +

	path	name	size	modificationTime
1	dbfs:/Volumes/project4_cat/project4_db/project4_vol/2019.csv	2019.csv	122681	1774582583000
2	dbfs:/Volumes/project4_cat/project4_db/project4_vol/2020.csv	2020.csv	283396	1774582584000
3	dbfs:/Volumes/project4_cat/project4_db/project4_vol/2021.csv	2021.csv	2869399	1774582584000

Reading the file for one year.

23 hours ago (2s) 2

```python
df = spark.read.csv("/Volumes/project4_cat/project4_db/project4_vol/2019.csv", header=True)
```

(1) Spark Jobs

df: pyspark.sql.dataframe.DataFrame = [SO43701: string, 11: string ... 7 more fields]

23 hours ago (1s) 3

```python
display(df);
```

(1) Spark Jobs

Table +

	SO43701	11	2019-07-01	Christy Zhu	christy12@adventure-works.com	Mountain-100 Silver, 44	16	3399.99	271.9992
1	SO43704	1	2019-07-01	Julio Ruiz	julio1@adventure-works.com	Mountain-100 Black, 48	1	3374.99	269.9992
2	SO43705	1	2019-07-01	Curtis Lu	curtis9@adventure-works.com	Mountain-100 Silver, 38	1	3399.99	271.9992
3	SO43700	1	2019-07-01	Ruben Prasad	ruben10@adventure-works.com	Road-650 Black, 62	1	699.0982	55.9279
4	SO43703	1	2019-07-01	Albert Alvarez	albert7@adventure-works.com	Road-150 Red, 62	1	3578.27	286.2616
5	SO43697	1	2019-07-01	Cole Watson	cole1@adventure-works.com	Road-150 Red, 62	1	3578.27	286.2616
6	SO43699	1	2019-07-01	Sydney Wright	sydney61@adventure-works.com	Mountain-100 Silver, 44	1	3399.99	271.9992
7	SO43702	1	2019-07-01	Colin Anand	colin45@adventure-works.com	Road-150 Red, 44	1	3578.27	286.2616
8	SO43698	1	2019-07-01	Rachael Martinez	rachael16@adventure-works.com	Mountain-100 Silver, 44	1	3399.99	271.9992
9	SO43707	1	2019-07-02	Emma Brown	emma3@adventure-works.com	Road-150 Red, 48	1	3578.27	286.2616
10	SO43711	1	2019-07-02	Courtney Edwards	courtney1@adventure-works.com	Road-150 Red, 56	1	3578.27	286.2616
11	SO43706	1	2019-07-02	Edward Brown	edward26@adventure-works.com	Road-150 Red, 48	1	3578.27	286.2616
12	SO43708	1	2019-07-02	Brad Deng	brad2@adventure-works.com	Road-650 Red, 52	1	699.0982	55.9279
13	SO43709	1	2019-07-02	Martha Xu	martha12@adventure-works.com	Road-150 Red, 52	1	3578.27	286.2616
14	SO43710	1	2019-07-02	Katrina Raji	katrina20@adventure-works.com	Road-150 Red, 56	1	3578.27	286.2616
15	SO43712	1	2019-07-02	Abigail Henderson	abigail73@adventure-works.com	Road-150 Red, 44	1	3578.27	286.2616

1,200 rows | 0.54s runtime

Bronze-Silver:

Changing the type of data using structype and strucfields.

```
from pyspark.sql.types import StructType, StructField, StringType, IntegerType, DoubleType, DateType

schema = StructType([                                   #StructType is a list of StructField objects and each StructField has a name, a type and an optional boolean flag to indicate whether the  field is nullable or not.
  StructField("SalesOrderID", StringType(), True),        #StructField represents a field in the schema
  StructField("CustomerID", IntegerType(), True),
  StructField("OrderDate", DateType(), True),
  StructField("CustomerName", StringType(), True),
  StructField("EmailAddress", StringType(), True),
  StructField("ProductName", StringType(), True),
  StructField("OrderQty", IntegerType(), True),
  StructField("UnitPrice", DoubleType(), True),
  StructField("TaxAmt", DoubleType(), True)
])
```

```
df= spark.read.format("csv").schema(schema).option("header", "true").load("/Volumes/project4_cat/project4_db/project4_vol/2019.csv")
df.show()
```

(1) Spark Jobs

df: pyspark.sql.dataframe.DataFrame = [SalesOrderID: string, CustomerID: integer ... 7 more fields]

SalesOrderID	CustomerID	OrderDate	CustomerName	EmailAddress	ProductName	OrderQty	UnitPrice	TaxAmt
SO43704	1	2019-07-01	Julio Ruiz	julio1@adventure-...	Mountain-100 Blac...	1	3374.99	269.9992
SO43705	1	2019-07-01	Curtis Lu	curtis9@adventure...	Mountain-100 Silv...	1	3399.99	271.9992
SO43700	1	2019-07-01	Ruben Prasad	ruben10@adventure...	Road-650 Black, 62	1	699.0982	55.9279
SO43703	1	2019-07-01	Albert Alvarez	albert7@adventure...	Road-150 Red, 62	1	3578.27	286.2616
SO43697	1	2019-07-01	Cole Watson	cole1@adventure-w...	Road-150 Red, 62	1	3578.27	286.2616
SO43699	1	2019-07-01	Sydney Wright	sydney61@adventur...	Mountain-100 Silv...	1	3399.99	271.9992
SO43702	1	2019-07-01	Colin Anand	colin45@adventure...	Road-150 Red, 44	1	3578.27	286.2616
SO43698	1	2019-07-01	Rachael Martinez	rachael16@adventu...	Mountain-100 Silv...	1	3399.99	271.9992
SO43707	1	2019-07-02	Emma Brown	emma3@adventure-w...	Road-150 Red, 48	1	3578.27	286.2616
SO43711	1	2019-07-02	Courtney Edwards	courtney1@adventu...	Road-150 Red, 56	1	3578.27	286.2616
SO43706	1	2019-07-02	Edward Brown	edward26@adventur...	Road-150 Red, 48	1	3578.27	286.2616
SO43708	1	2019-07-02	Brad Deng	brad2@adventure-w...	Road-650 Red, 52	1	699.0982	55.9279
SO43709	1	2019-07-02	Martha Xu	martha12@adventur...	Road-150 Red, 52	1	3578.27	286.2616

Reading the data using spark.read.

The Show function brings the data in text format while the display function brings it in UI format.

```
df.display()
```

(1) Spark Jobs

Table

	SalesOrderID	CustomerID	OrderDate	CustomerName	EmailAddress	ProductName	OrderQty	UnitPrice	TaxAmt
11	SO43706	1	2019-07-02	Edward Brown	edward26@adventure-works.com	Road-150 Red, 48	1	3578.27	286.2616
12	SO43708	1	2019-07-02	Brad Deng	brad2@adventure-works.com	Road-650 Red, 52	1	699.0982	55.9279
13	SO43709	1	2019-07-02	Martha Xu	martha12@adventure-works.com	Road-150 Red, 52	1	3578.27	286.2616
14	SO43710	1	2019-07-02	Katrina Raji	katrina20@adventure-works.com	Road-150 Red, 56	1	3578.27	286.2616
15	SO43712	1	2019-07-02	Abigail Henderson	abigail73@adventure-works.com	Road-150 Red, 44	1	3578.27	286.2616
16	SO43720	1	2019-07-03	Melanie Sanchez	melanie47@adventure-works.com	Road-150 Red, 44	1	3578.27	286.2616
17	SO43721	1	2019-07-03	Louis Xie	louis20@adventure-works.com	Road-150 Red, 62	1	3578.27	286.2616
18	SO43714	1	2019-07-03	Latasha Alonso	latasha8@adventure-works.com	Road-150 Red, 44	1	3578.27	286.2616
19	SO43715	1	2019-07-03	Warren Jai	warren42@adventure-works.com	Road-150 Red, 56	1	3578.27	286.2616
20	SO43716	1	2019-07-03	Jonathon Gutierrez	jonathon8@adventure-works.com	Road-150 Red, 44	1	3578.27	286.2616
21	SO43719	1	2019-07-03	Lucas Hill	lucas45@adventure-works.com	Road-150 Red, 48	1	3578.27	286.2616
22	SO43713	1	2019-07-03	Sydney Rogers	sydney3@adventure-works.com	Road-150 Red, 62	1	3578.27	286.2616
23	SO43717	1	2019-07-03	Alexandra Watson	alexandra20@adventure-works.com	Road-650 Black, 62	1	699.0982	55.9279
24	SO43718	1	2019-07-03	Kevin Gonzalez	kevin46@adventure-works.com	Road-150 Red, 44	1	3578.27	286.2616
25	SO43727	1	2019-07-04	Logan Gonzales	logan18@adventure-works.com	Road-150 Red, 44	1	3578.27	286.2616

1,200 rows | 0.29s runtime

```
df=df.drop(col("CustomerID"))
```

df: pyspark.sql.dataframe.DataFrame = [SalesOrderID: string, OrderDate: date ... 6 more fields]

Counting the rows.

Gold | Silver ×

File Edit View Run Help Python Tabs: ON Last edit was 2 days ago

2 days ago (<1s) 5

```
df.display()
```

(1) Spark Jobs

Table

	SalesOrderID	OrderDate	CustomerName	EmailAddress	ProductName	OrderQty	UnitPrice	TaxAmt
1	SO43704	2019-07-01	Julio Ruiz	julio1@adventure-works.com	Mountain-100 Black, 48	1	3374.99	269.9992
2	SO43705	2019-07-01	Curtis Lu	curtis9@adventure-works.com	Mountain-100 Silver, 38	1	3399.99	271.9992
3	SO43700	2019-07-01	Ruben Prasad	ruben10@adventure-works.com	Road-650 Black, 62	1	699.0982	55.9279
4	SO43703	2019-07-01	Albert Alvarez	albert7@adventure-works.com	Road-150 Red, 62	1	3578.27	286.2616
5	SO43697	2019-07-01	Cole Watson	cole1@adventure-works.com	Road-150 Red, 62	1	3578.27	286.2616
6	SO43699	2019-07-01	Sydney Wright	sydney61@adventure-works.com	Mountain-100 Silver, 44	1	3399.99	271.9992
7	SO43702	2019-07-01	Colin Anand	colin45@adventure-works.com	Road-150 Red, 44	1	3578.27	286.2616
8	SO43698	2019-07-01	Rachael Martinez	rachael16@adventure-works.com	Mountain-100 Silver, 44	1	3399.99	271.9992
9	SO43707	2019-07-02	Emma Brown	emma3@adventure-works.com	Road-150 Red, 48	1	3578.27	286.2616
10	SO43711	2019-07-02	Courtney Edwards	courtney1@adventure-works.com	Road-150 Red, 56	1	3578.27	286.2616
11	SO43706	2019-07-02	Edward Brown	edward26@adventure-works.com	Road-150 Red, 48	1	3578.27	286.2616
12	SO43708	2019-07-02	Brad Deng	brad2@adventure-works.com	Road-650 Red, 52	1	699.0982	55.9279
13	SO43709	2019-07-02	Martha Xu	martha12@adventure-works.com	Road-150 Red, 52	1	3578.27	286.2616
14	SO43710	2019-07-02	Katrina Raji	katrina20@adventure-works.com	Road-150 Red, 56	1	3578.27	286.2616
15	SO43712	2019-07-02	Abigail Henderson	abigail73@adventure-works.com	Road-150 Red, 44	1	3578.27	286.2616

1,200 rows | 0.31s runtime

2 days ago (<1s) 6

```
df.count()
```

(2) Spark Jobs

```
1200
```

Checking for null values and trimming the columns.

Gold | Silver ×

File Edit View Run Help Python Tabs: ON Last edit was 2 days ago

2 days ago (<1s) 7

```
from pyspark.sql.functions import *
```

2 days ago (1s) 8

```
from pyspark.sql.functions import col, sum

df.select([
    sum(col(c).isNull().cast("int")).alias(c)
    for c in df.columns
]).display()
```

(2) Spark Jobs

Table

	SalesOrderID	OrderDate	CustomerName	EmailAddress	ProductName	OrderQty	UnitPrice	TaxAmt
1	0	0	0	0	0	0	0	0

1 row | 0.65s runtime

2 days ago (<1s) 9

```
from pyspark.sql.functions import upper, trim
df = df.withColumn("CustomerName", trim(upper(col("CustomerName"))))
df = df.withColumn("EmailAddress", trim(upper(col("EmailAddress"))))
df = df.withColumn("ProductName", trim(upper(col("ProductName"))))
```

df: pyspark.sql.dataframe.DataFrame = [SalesOrderID: string, OrderDate: date ... 6 more fields]

2 days ago (<1s) 10

```
df.display()
```

(1) Spark Jobs

Checking for duplicates using window and row number.

Gold | Silver

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2 days ago (<1s) 11

```
df=df.filter(
    col("SalesOrderID").isNotNull() &
    col("EmailAddress").isNotNull() &
    col("ProductName").isNotNull())
```

df: pyspark.sql.dataframe.DataFrame = [SalesOrderID: string, OrderDate: date ... 6 more fields]

2 days ago (<1s) 12

```
from pyspark.sql.window import Window
from pyspark.sql.functions import row_number

window_spec = Window.partitionBy("SalesOrderID").orderBy(col("OrderDate").desc())

df = df.withColumn("rn", row_number().over(window_spec)) \
      .filter(col("rn") == 1) \
      .drop("rn")
```

df: pyspark.sql.dataframe.DataFrame = [SalesOrderID: string, OrderDate: date ... 6 more fields]

2 days ago (1s) 13

```
df.display()
```

(2) Spark Jobs

Table

	SalesOrderID	OrderDate	CustomerName	EmailAddress	ProductName	OrderQty	UnitPrice	TaxAmt
1	SO43697	2019-07-01	COLE WATSON	COLE1@ADVENTURE-WORKS.COM	ROAD-150 RED, 62	1	3578.27	286.2616
2	SO43698	2019-07-01	RACHAEL MARTINEZ	RACHAEL16@ADVENTURE-WORKS.COM	MOUNTAIN-100 SILVER, 44	1	3399.99	271.9992
3	SO43699	2019-07-01	SYDNEY WRIGHT	SYDNEY61@ADVENTURE-WORKS.COM	MOUNTAIN-100 SILVER, 44	1	3399.99	271.9992
4	SO43700	2019-07-01	RUBEN PRASAD	RUBEN10@ADVENTURE-WORKS.COM	ROAD-650 BLACK, 62	1	699.0982	55.9279
5	SO43702	2019-07-01	COLIN ANAND	COLIN45@ADVENTURE-WORKS.COM	ROAD-150 RED, 44	1	3578.27	286.2616
6	SO43703	2019-07-01	ALBERT ALVAREZ	ALBERT7@ADVENTURE-WORKS.COM	ROAD-150 RED, 62	1	3578.27	286.2616
7	SO43704	2019-07-01	JULIO RUIZ	JULIO1@ADVENTURE-WORKS.COM	MOUNTAIN-100 BLACK, 48	1	3374.99	269.9992
8	SO43705	2019-07-01	CURTIS LU	CURTIS9@ADVENTURE-WORKS.COM	MOUNTAIN-100 SILVER, 38	1	3399.99	271.9992
9	SO43706	2019-07-02	EDWARD BROWN	EDWARD26@ADVENTURE-WORKS.COM	ROAD-150 RED, 48	1	3578.27	286.2616
10	SO43707	2019-07-02	EMMA BROWN	EMMA3@ADVENTURE-WORKS.COM	ROAD-150 RED, 48	1	3578.27	286.2616
11	SO43708	2019-07-02	BRAD DENG	BRAD2@ADVENTURE-WORKS.COM	ROAD-650 RED, 52	1	699.0982	55.9279

Filtering the data applying conditions.

2 days ago (<1s) 14

```
df = df.filter(col("TaxAmt") >=65)
```

df: pyspark.sql.dataframe.DataFrame = [SalesOrderID: string, OrderDate: date ... 6 more fields]

2 days ago (<1s) 15

```
df.display()
```

(2) Spark Jobs

Table

	SalesOrderID	OrderDate	CustomerName	EmailAddress	ProductName	OrderQty	UnitPrice	TaxAmt
1	SO43697	2019-07-01	COLE WATSON	COLE1@ADVENTURE-WORKS.COM	ROAD-150 RED, 62	1	3578.27	286.2616
2	SO43698	2019-07-01	RACHAEL MARTINEZ	RACHAEL16@ADVENTURE-WORKS.COM	MOUNTAIN-100 SILVER, 44	1	3399.99	271.9992
3	SO43699	2019-07-01	SYDNEY WRIGHT	SYDNEY61@ADVENTURE-WORKS.COM	MOUNTAIN-100 SILVER, 44	1	3399.99	271.9992
4	SO43702	2019-07-01	COLIN ANAND	COLIN45@ADVENTURE-WORKS.COM	ROAD-150 RED, 44	1	3578.27	286.2616
5	SO43703	2019-07-01	ALBERT ALVAREZ	ALBERT7@ADVENTURE-WORKS.COM	ROAD-150 RED, 62	1	3578.27	286.2616
6	SO43704	2019-07-01	JULIO RUIZ	JULIO1@ADVENTURE-WORKS.COM	MOUNTAIN-100 BLACK, 48	1	3374.99	269.9992
7	SO43705	2019-07-01	CURTIS LU	CURTIS9@ADVENTURE-WORKS.COM	MOUNTAIN-100 SILVER, 38	1	3399.99	271.9992
8	SO43706	2019-07-02	EDWARD BROWN	EDWARD26@ADVENTURE-WORKS.COM	ROAD-150 RED, 48	1	3578.27	286.2616
9	SO43707	2019-07-02	EMMA BROWN	EMMA3@ADVENTURE-WORKS.COM	ROAD-150 RED, 48	1	3578.27	286.2616
10	SO43709	2019-07-02	MARTHA XU	MARTHA12@ADVENTURE-WORKS.COM	ROAD-150 RED, 52	1	3578.27	286.2616
11	SO43710	2019-07-02	KATRINA RAJI	KATRINA20@ADVENTURE-WORKS.COM	ROAD-150 RED, 56	1	3578.27	286.2616
12	SO43711	2019-07-02	COURTNEY EDWARDS	COURTNEY1@ADVENTURE-WORKS.COM	ROAD-150 RED, 56	1	3578.27	286.2616
13	SO43712	2019-07-02	ABIGAIL HENDERSON	ABIGAIL73@ADVENTURE-WORKS.COM	ROAD-150 RED, 44	1	3578.27	286.2616
14	SO43713	2019-07-03	SYDNEY ROGERS	SYDNEY3@ADVENTURE-WORKS.COM	ROAD-150 RED, 62	1	3578.27	286.2616
15	SO43714	2019-07-03	LATASHA ALONSO	LATASHA8@ADVENTURE-WORKS.COM	ROAD-150 RED, 44	1	3578.27	286.2616

1,063 rows | 0.47s runtime

Saving a delta table.

2 days ago (1s)

```
%sql
create schema if not exists project4_cat.silver
```

OK

2 days ago (11s)

```
df.write.format("delta")\
    .mode("overwrite")\
.saveAsTable("project4_cat.silver.sales")
```

(2) Spark Jobs

Silver-Gold:

Reading the table from the silver layer.

Gold × Silver +

File Edit View Run Help Python Tabs: ON Last edit was 2 days ago

2 days ago (<1s) 1

```
df=spark.table("project4_cat.silver.sales")
```

df: pyspark.sql.dataframe.DataFrame = [SalesOrderID: string, OrderDate: date ... 6 more fields]

2 days ago (2s) 2

```
df.display()
```

(2) Spark Jobs

Table +

	SalesOrderID	OrderDate	CustomerName	EmailAddress	ProductName	OrderQty	UnitPrice	TaxAmt
1	SO43697	2019-07-01	COLE WATSON	COLE1@ADVENTURE-WORKS.COM	ROAD-150 RED, 62	1	3578.27	286.2616
2	SO43698	2019-07-01	RACHAEL MARTINEZ	RACHAEL16@ADVENTURE-WORKS.COM	MOUNTAIN-100 SILVER, 44	1	3399.99	271.9992
3	SO43699	2019-07-01	SYDNEY WRIGHT	SYDNEY61@ADVENTURE-WORKS.COM	MOUNTAIN-100 SILVER, 44	1	3399.99	271.9992
4	SO43702	2019-07-01	COLIN ANAND	COLIN45@ADVENTURE-WORKS.COM	ROAD-150 RED, 44	1	3578.27	286.2616
5	SO43703	2019-07-01	ALBERT ALVAREZ	ALBERT7@ADVENTURE-WORKS.COM	ROAD-150 RED, 62	1	3578.27	286.2616
6	SO43704	2019-07-01	JULIO RUIZ	JULIO1@ADVENTURE-WORKS.COM	MOUNTAIN-100 BLACK, 48	1	3374.99	269.9992
7	SO43705	2019-07-01	CURTIS LU	CURTIS9@ADVENTURE-WORKS.COM	MOUNTAIN-100 SILVER, 38	1	3399.99	271.9992
8	SO43706	2019-07-02	EDWARD BROWN	EDWARD26@ADVENTURE-WORKS.COM	ROAD-150 RED, 48	1	3578.27	286.2616
9	SO43707	2019-07-02	EMMA BROWN	EMMA3@ADVENTURE-WORKS.COM	ROAD-150 RED, 48	1	3578.27	286.2616
10	SO43709	2019-07-02	MARTHA XU	MARTHA12@ADVENTURE-WORKS.COM	ROAD-150 RED, 52	1	3578.27	286.2616
11	SO43710	2019-07-02	KATRINA RAJI	KATRINA20@ADVENTURE-WORKS.COM	ROAD-150 RED, 56	1	3578.27	286.2616
12	SO43711	2019-07-02	COURTNEY EDWARDS	COURTNEY1@ADVENTURE-WORKS.COM	ROAD-150 RED, 56	1	3578.27	286.2616
13	SO43712	2019-07-02	ABIGAIL HENDERSON	ABIGAIL73@ADVENTURE-WORKS.COM	ROAD-150 RED, 44	1	3578.27	286.2616
14	SO43713	2019-07-03	SYDNEY ROGERS	SYDNEY3@ADVENTURE-WORKS.COM	ROAD-150 RED, 62	1	3578.27	286.2616
15	SO43714	2019-07-03	LATASHA ALONSO	LATASHA8@ADVENTURE-WORKS.COM	ROAD-150 RED, 44	1	3578.27	286.2616

1,063 rows | 2.04s runtime

Performing aggregations.

Total Revenue by Product

```
from pyspark.sql.functions import col, sum  df_gold_product_revenue = df.withCol ...
```

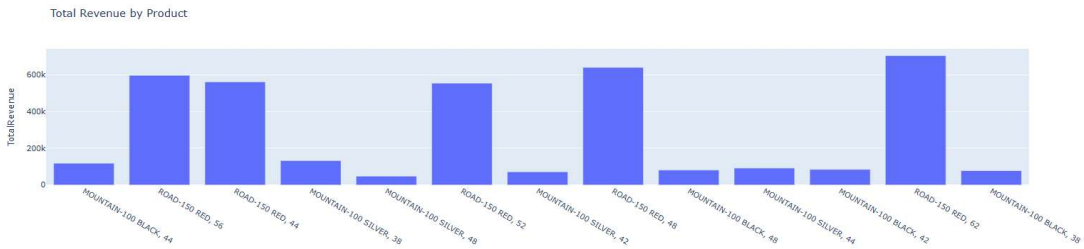
```
import plotly.express as px

pdf = df_gold_product_revenue.toPandas()

fig = px.bar(pdf, x="ProductName", y="TotalRevenue",
             title="Total Revenue by Product")
fig.show()
```

(2) Spark Jobs

pdf: pandas.core.frame.DataFrame = [ProductName: object, TotalRevenue: float64]



Revenue Trend Over Time

```
df_gold_daily_sales = df.withColumn(
    "Revenue", col("OrderQty") * col("UnitPrice")
).groupBy("OrderDate").agg(
    sum("Revenue").alias("DailyRevenue")
).orderBy("OrderDate")
```

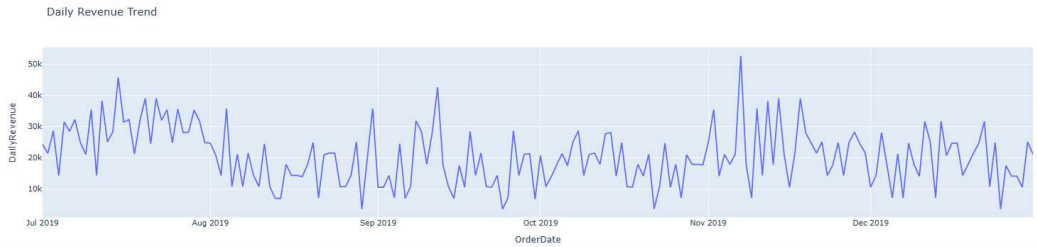
df_gold_daily_sales: pyspark.sql.dataframe.DataFrame = [OrderDate: date, DailyRevenue: double]

```
pdf = df_gold_daily_sales.toPandas()

fig = px.line(pdf, x="OrderDate", y="DailyRevenue",
              title="Daily Revenue Trend")
fig.show()
```

(4) Spark Jobs

pdf: pandas.core.frame.DataFrame = [OrderDate: object, DailyRevenue: float64]



Top Customers by Revenue

```
df_gold_top_customers = df.withColumn(
    "Revenue", col("OrderQty") * col("UnitPrice")
).groupBy("CustomerName").agg(
    sum("Revenue").alias("TotalRevenue")
).orderBy(col("TotalRevenue").desc())
```

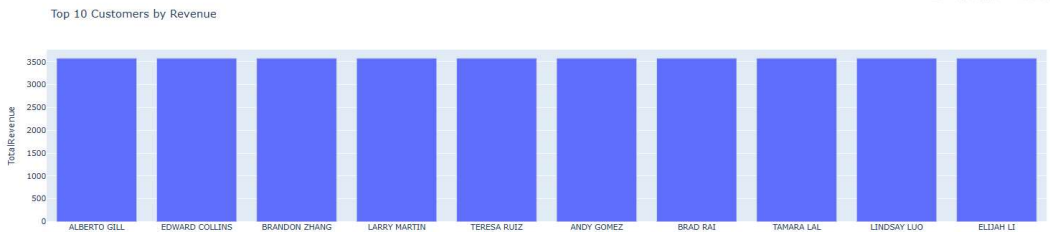
df_gold_top_customers: pyspark.sql.dataframe.DataFrame = [CustomerName: string, TotalRevenue: double]

```
pdf = df_gold_top_customers.limit(10).toPandas()

fig = px.bar(pdf, x="CustomerName", y="TotalRevenue",
             title="Top 10 Customers by Revenue")
fig.show()
```

(2) Spark Jobs

pdf: pandas.core.frame.DataFrame = [CustomerName: object, TotalRevenue: float64]



Average Order Value (AOV)

```
from pyspark.sql.functions import avg

df_order_value = df.withColumn(
    "Revenue", col("OrderQty") * col("UnitPrice")
).groupBy("SalesOrderID").agg(
    sum("Revenue").alias("OrderValue")
)

df_gold_aov = df_order_value.agg(
    avg("OrderValue").alias("AvgOrderValue")
)
```

df_gold_aov: pyspark.sql.dataframe.DataFrame = [AvgOrderValue: double]
df_order_value: pyspark.sql.dataframe.DataFrame = [SalesOrderID: string, OrderValue: double]

```
pdf = df_order_value.toPandas()

fig = px.histogram(pdf, x="OrderValue",
                   title="Distribution of Order Values")
fig.show()
```

(2) Spark Jobs

pdf: pandas.core.frame.DataFrame = [SalesOrderID: object, OrderValue: float64]

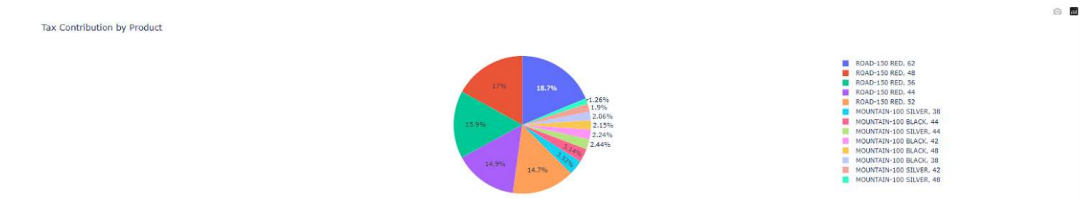
Tax Contribution by Product

```
df_gold_tax = df.groupBy("ProductName").agg(
    sum("TaxAmt").alias("TotalTax")
)
```

df_gold_tax: pyspark.sql.dataframe.DataFrame = [ProductName: string, TotalTax: double]

```
pdf = df_gold_tax.toPandas()

fig = px.pie(pdf, names="ProductName", values="TotalTax",
             title="Tax Contribution by Product")
fig.show()
```

(2) Spark Jobs

pdf: pandas.core.frame.DataFrame = [ProductName: object, TotalTax: float64]